



#### INCLUDES

- ✓ Course framework
- ✓ Instructional section
- ✓ Sample exam questions

# AP<sup>®</sup> Physics 1: Algebra-Based

COURSE AND EXAM DESCRIPTION

Effective  
**Fall 2024**

# ADVANCED PLACEMENT PHYSICS 1 TABLE OF INFORMATION

## CONSTANTS AND CONVERSION FACTORS

|                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Universal gravitational constant,<br>$G = 6.67 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2) = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$<br>1 atmosphere of pressure,<br>$1 \text{ atm} = 1.0 \times 10^5 \text{ N/m}^2 = 1.0 \times 10^5 \text{ Pa}$ | Magnitude of the acceleration due to gravity at<br>Earth's surface, $g = 9.8 \text{ m/s}^2$<br>Magnitude of the gravitational field strength at<br>Earth's surface, $g = 9.8 \text{ N/kg}$ |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## PREFIXES

| Factor     | Prefix | Symbol |
|------------|--------|--------|
| $10^{12}$  | tera   | T      |
| $10^9$     | giga   | G      |
| $10^6$     | mega   | M      |
| $10^3$     | kilo   | k      |
| $10^{-2}$  | centi  | c      |
| $10^{-3}$  | milli  | m      |
| $10^{-6}$  | micro  | $\mu$  |
| $10^{-9}$  | nano   | n      |
| $10^{-12}$ | pico   | p      |

## UNIT SYMBOLS

|           |    |         |    |
|-----------|----|---------|----|
| hertz,    | Hz | newton, | N  |
| joule,    | J  | pascal, | Pa |
| kilogram, | kg | second, | s  |
| meter,    | m  | watt,   | W  |

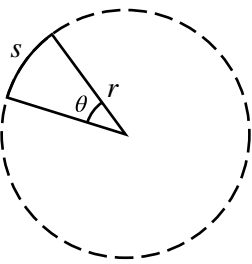
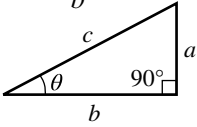
## VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES

| $\theta$      | $0^\circ$ | $30^\circ$   | $37^\circ$ | $45^\circ$   | $53^\circ$ | $60^\circ$   | $90^\circ$ |
|---------------|-----------|--------------|------------|--------------|------------|--------------|------------|
| $\sin \theta$ | 0         | $1/2$        | $3/5$      | $\sqrt{2}/2$ | $4/5$      | $\sqrt{3}/2$ | 1          |
| $\cos \theta$ | 1         | $\sqrt{3}/2$ | $4/5$      | $\sqrt{2}/2$ | $3/5$      | $1/2$        | 0          |
| $\tan \theta$ | 0         | $\sqrt{3}/3$ | $3/4$      | 1            | $4/3$      | $\sqrt{3}$   | $\infty$   |

The following conventions are used in this exam:

- The frame of reference of any problem is assumed to be inertial unless otherwise stated.
- Air resistance is assumed to be negligible unless otherwise stated.
- Springs and strings are assumed to be ideal unless otherwise stated.
- Fluids are assumed to be ideal, and pipes are assumed to be completely filled by fluid, unless otherwise stated.

## GEOMETRY AND TRIGONOMETRY

|                                                          |                                                                |                                                                                     |                                                                                                                   |
|----------------------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Rectangle<br>$A = bh$                                    | Rectangular Solid<br>$V = \ell wh$                             |  | Right Triangle<br>$a^2 + b^2 = c^2$                                                                               |
| Triangle<br>$A = \frac{1}{2}bh$                          | Cylinder<br>$V = \pi r^2 \ell$<br>$S = 2\pi r \ell + 2\pi r^2$ |                                                                                     | $\sin \theta = \frac{a}{c}$                                                                                       |
| Circle<br>$A = \pi r^2$<br>$C = 2\pi r$<br>$s = r\theta$ | Sphere<br>$V = \frac{4}{3}\pi r^3$<br>$S = 4\pi r^2$           |                                                                                     | $\cos \theta = \frac{b}{c}$                                                                                       |
|                                                          |                                                                |                                                                                     | $\tan \theta = \frac{a}{b}$  |

# MECHANICS AND FLUIDS

$$v_x = v_{x0} + a_x t$$

$$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{x0}^2 + 2a_x (x - x_0)$$

$$\vec{x}_{\text{cm}} = \frac{\sum m_i \vec{x}_i}{\sum m_i}$$

$$\vec{a}_{\text{sys}} = \frac{\sum \vec{F}}{m_{\text{sys}}} = \frac{\vec{F}_{\text{net}}}{m_{\text{sys}}}$$

$$|\vec{F}_g| = G \frac{m_1 m_2}{r^2}$$

$$|\vec{F}_f| \leq |\mu \vec{F}_n|$$

$$\vec{F}_s = -k \Delta \vec{x}$$

$$a_c = \frac{v^2}{r}$$

$$K = \frac{1}{2} m v^2$$

$$W = F_{\parallel} d = F d \cos \theta$$

$$\Delta K = \sum W_i = \sum F_{\parallel, i} d_i$$

$$U_s = \frac{1}{2} k (\Delta x)^2$$

$$U_G = -\frac{G m_1 m_2}{r}$$

$$\Delta U_g = m g \Delta y$$

$$P_{\text{avg}} = \frac{W}{\Delta t} = \frac{\Delta E}{\Delta t}$$

$$P_{\text{inst}} = F_{\parallel} v = F v \cos \theta$$

$$\vec{p} = m \vec{v}$$

$$\vec{F}_{\text{net}} = \frac{\Delta \vec{p}}{\Delta t} = m \frac{\Delta \vec{v}}{\Delta t} = m \vec{a}$$

$$\vec{J} = \vec{F}_{\text{avg}} \Delta t = \Delta \vec{p}$$

$$\vec{v}_{\text{cm}} = \frac{\sum \vec{p}_i}{\sum m_i} = \frac{\sum m_i \vec{v}_i}{\sum m_i}$$

$a$  = acceleration

$d$  = distance

$E$  = energy

$F$  = force

$J$  = impulse

$k$  = spring constant

$K$  = kinetic energy

$m$  = mass

$p$  = momentum

$P$  = power

$r$  = radius, distance, or position

$t$  = time

$U$  = potential energy

$v$  = velocity or speed

$W$  = work

$x$  = position

$y$  = height

$\theta$  = angle

$\mu$  = coefficient of friction

$$\omega = \omega_0 + \alpha t$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

$$v = r\omega$$

$$a_r = r\alpha$$

$$\tau = r_{\perp} F = r F \sin \theta$$

$$I = \sum m_i r_i^2$$

$$I' = I_{\text{cm}} + M d^2$$

$$\alpha_{\text{sys}} = \frac{\sum \tau}{I_{\text{sys}}} = \frac{\tau_{\text{net}}}{I_{\text{sys}}}$$

$$K = \frac{1}{2} I \omega^2$$

$$W = \tau \Delta \theta$$

$$L = I \omega$$

$$L = r m v \sin \theta$$

$$\Delta L = \tau \Delta t$$

$$\Delta x_{\text{cm}} = r \Delta \theta$$

$$T = \frac{1}{f}$$

$$T_s = 2\pi \sqrt{\frac{m}{k}}$$

$$T_p = 2\pi \sqrt{\frac{\ell}{g}}$$

$$x = A \cos(2\pi f t)$$

$$x = A \sin(2\pi f t)$$

$$\rho = \frac{m}{V}$$

$$P = \frac{F_{\perp}}{A}$$

$$P = P_0 + \rho g h$$

$$P_{\text{gauge}} = \rho g h$$

$$F_b = \rho V g$$

$$A_1 v_1 = A_2 v_2$$

$$P_1 + \rho g y_1 + \frac{1}{2} \rho v_1^2 = P_2 + \rho g y_2 + \frac{1}{2} \rho v_2^2$$

$a$  = acceleration

$A$  = amplitude or area

$d$  = distance

$f$  = frequency

$F$  = force

$h$  = height

$I$  = rotational inertia

$k$  = spring constant

$K$  = kinetic energy

$\ell$  = length

$L$  = angular momentum

$m$  = mass

$M$  = mass

$P$  = pressure

$r$  = radius, distance, or position

$t$  = time

$T$  = period

$v$  = velocity or speed

$V$  = volume

$W$  = work

$x$  = position

$y$  = vertical position

$\alpha$  = angular acceleration

$\theta$  = angle

$\rho$  = density

$\tau$  = torque

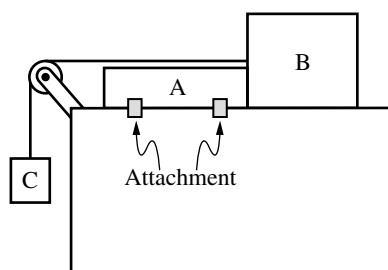
$\omega$  = angular speed

# Sample Exam Questions

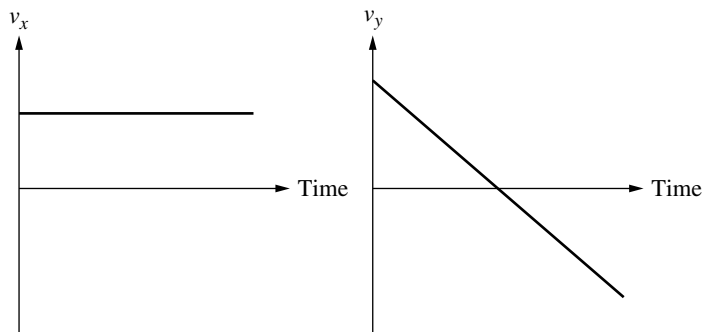
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The sample exam questions that follow illustrate the relationship between the course framework and the AP Physics 1 Exam and serve as examples of the types of questions that appear on the exam. These sample questions do not represent the full range and distribution of items on an official AP Physics 1 Exam. After the sample questions is a table which shows which skill, learning objective, and essential knowledge statement each question relates to. This table also provides the answers to the multiple-choice questions.

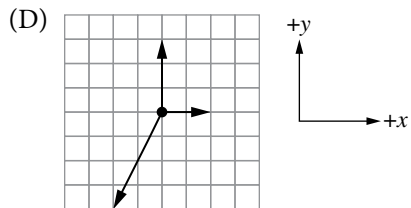
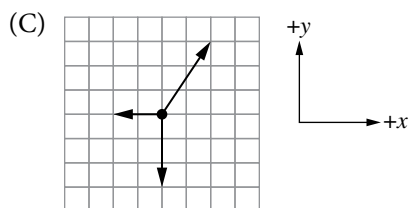
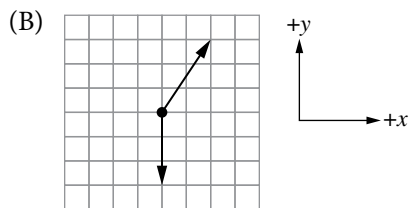
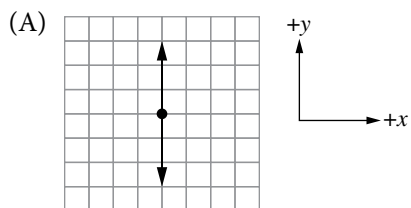
## Section I: Multiple-Choice Questions

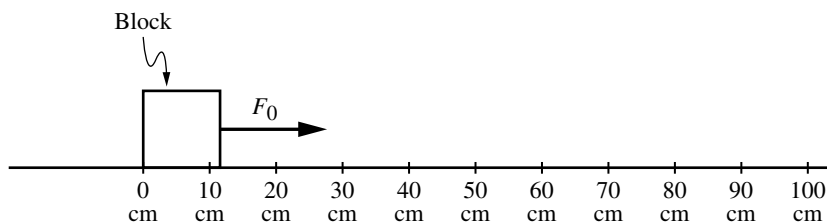


1. Blocks A and B, of masses  $m_A$  and  $m_B$ , respectively, are at rest on a horizontal surface with negligible friction. Block A is attached to the table. Block C of mass  $m_C$  is suspended by a string that is tied to Block B and passes over a pulley as shown. The blocks all remain at rest. Which of the following gives the magnitude of the force exerted by Block A on Block B?
- (A)  $m_B g$
- (B)  $m_C g$
- (C)  $\frac{m_A m_C g}{m_A + m_B}$
- (D)  $\frac{m_B m_C g}{m_A + m_B}$



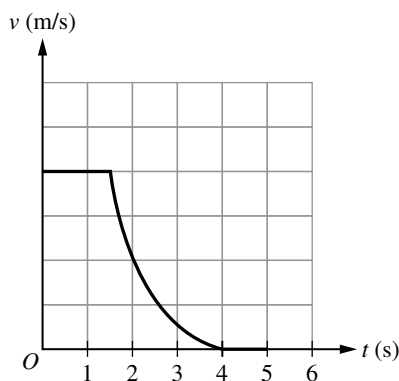
2. An object is subject to multiple forces that result in the object having horizontal and vertical velocity components  $v_x$  and  $v_y$ , respectively as a function of time, as shown. Which of the following diagrams could represent the forces exerted on the object?





3. A constant force  $F_0$  is exerted continuously on a block that is initially at rest on a horizontal surface. The change in kinetic energy of the block while the block is moved from position  $x = 0$  to  $x = 50$  cm is  $\Delta K_1$ , and the change in kinetic energy of the block as the block is moved from  $x = 50$  cm to  $x = 100$  cm is  $\Delta K_2$ . How does  $\Delta K_1$  compare to  $\Delta K_2$ , and why?
- (A)  $\Delta K_2 = \Delta K_1$  because the velocity increases by the same amount between 0 cm to 50 cm and between 50 cm to 100 cm.
- (B)  $\Delta K_2 = \Delta K_1$  because the applied force does the same work on the block between 0 cm to 50 cm and between 50 cm to 100 cm.
- (C)  $\Delta K_2 > \Delta K_1$  because the block is moving faster on average between 50 cm to 100 cm than it is between 0 cm to 50 cm.
- (D)  $\Delta K_2 > \Delta K_1$  because the rate of change of kinetic energy is greater between 50 cm to 100 cm than it is between 0 cm to 50 cm.

Questions 4 and 5 refer to the following.

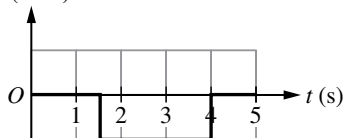


At time  $t = 0$ , a car is moving in the positive direction along a straight road. The graph represents the car's velocity  $v$  as a function of time  $t$ . The car's position at  $t = 0$  is  $x = 0$ .

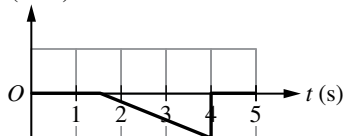
4. The position of the car at times 1 s, 2 s, and 3 s are  $x_1$ ,  $x_2$ , and  $x_3$ , respectively. Which of the following correctly ranks  $x_1$ ,  $x_2$ , and  $x_3$  from greatest to least?
- (A)  $x_1 > x_2 > x_3$
- (B)  $x_3 > x_2 > x_1$
- (C)  $x_2 > x_3 > x_1$
- (D)  $x_1 > x_3 > x_2$

5. Which of the following graphs could represent the acceleration  $a$  of the car as a function of time  $t$ ?

(A)  $a$  (m/s<sup>2</sup>)



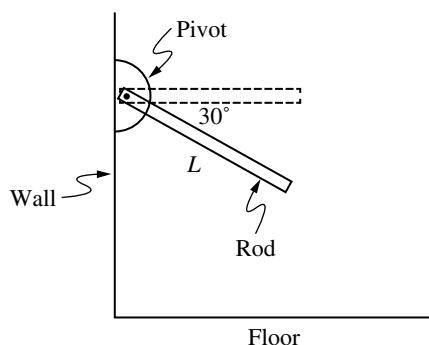
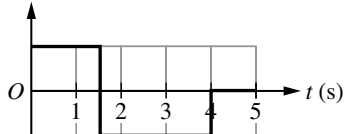
(B)  $a$  (m/s<sup>2</sup>)



(C)  $a$  (m/s<sup>2</sup>)



(D)  $a$  (m/s<sup>2</sup>)



6. A uniform rod of mass  $M_0$  and length  $L$  is free to rotate about a pivot at its left end and is released from rest when the rod is  $30^\circ$  below the horizontal, as shown in the figure. With respect to the pivot, the rod has rotational inertia  $I_0 = \frac{1}{3} M_0 L^2$ .

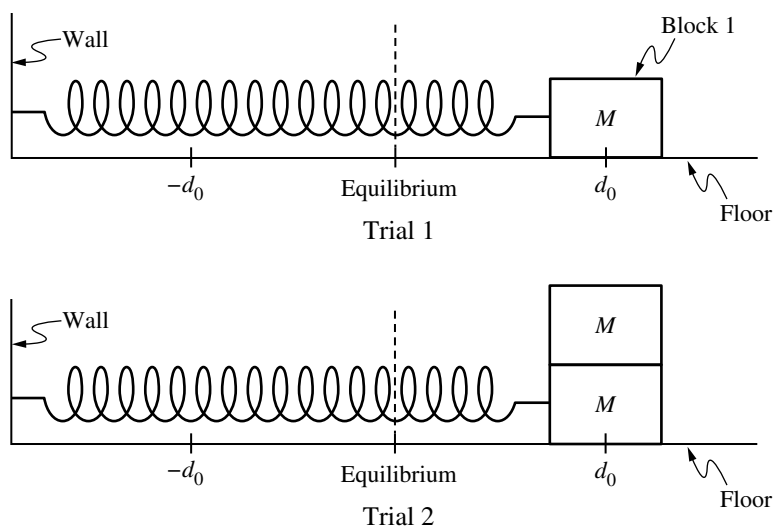
Which of the following expressions correctly represents the magnitude of the net torque exerted on the rod about the pivot at the moment the rod is released?

(A)  $M_0 g \left( \frac{L}{2} \right) \sin(30^\circ)$

(B)  $M_0 g \left( \frac{L}{2} \right) \sin(60^\circ)$

(C)  $M_0 g L \sin(30^\circ)$

(D)  $M_0 g L \sin(60^\circ)$



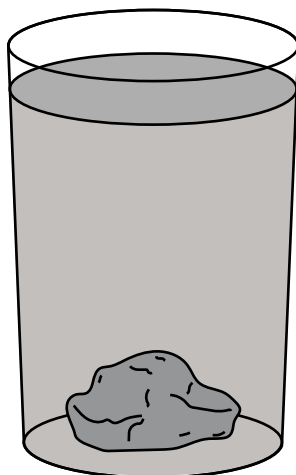
7. In Trial 1 of an experiment, Block 1 of mass  $M$  is oscillating at the end of a spring. The other end of the spring is attached to a wall. Friction between the block and the floor is negligible. Block 1 oscillates with amplitude  $d_0$ .
- In Trial 2 of the experiment, a second identical block is attached to Block 1, and the two-block system oscillates with amplitude  $d_0$ .

How do the maximum speeds and maximum kinetic energies in the two trials compare?

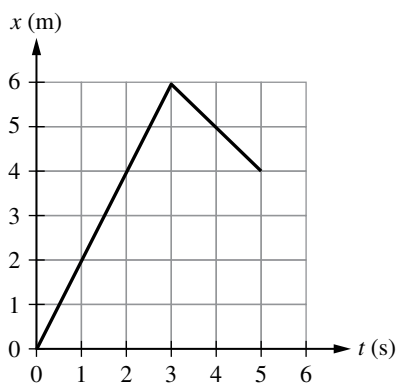
| Maximum Speed            | Maximum Kinetic Energy |
|--------------------------|------------------------|
| (A) Greater in Trial 1   | Greater in Trial 1     |
| (B) Greater in Trial 1   | Equal in both trials   |
| (C) Equal in both trials | Equal in both trials   |
| (D) Equal in both trials | Greater in Trial 2     |

(A) A  
(B) B  
(C) C  
(D) D

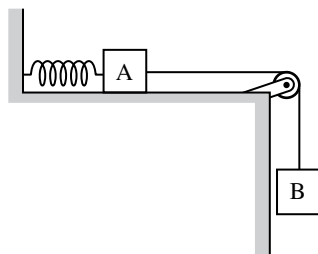




8. A small rock sits at the bottom of a cup filled with water. The upward force exerted by the water on the rock is  $F_0$ . The water is then poured out and replaced by an oil that is  $\frac{3}{4}$  as dense as water, and the rock again sits at the bottom of the cup, completely under the oil. Which of the following expressions correctly represents the magnitude of the upward force exerted by the oil on the rock?
- (A)  $\frac{3}{4}F_0$
- (B)  $F_0$
- (C)  $\frac{4}{3}F_0$
- (D) The answer cannot be determined without knowing the rock's volume.



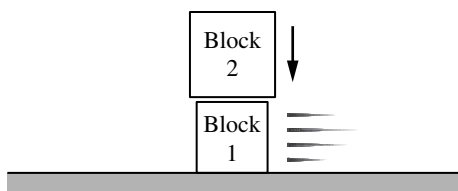
9. Two blocks move on a straight, level track, where friction between the blocks and the track is negligible. Block B of mass 4.0 kg is initially at rest. Block A of mass 2.0 kg slides to the right and collides with Block B at time  $t = 3$  s. The figure shows the position  $x$  of Block A as a function of time  $t$ . Which of the following is most nearly the momentum of Block B after the collision?
- (A) 2.0 kg·m/s
- (B) 3.0 kg·m/s
- (C) 4.0 kg·m/s
- (D) 6.0 kg·m/s



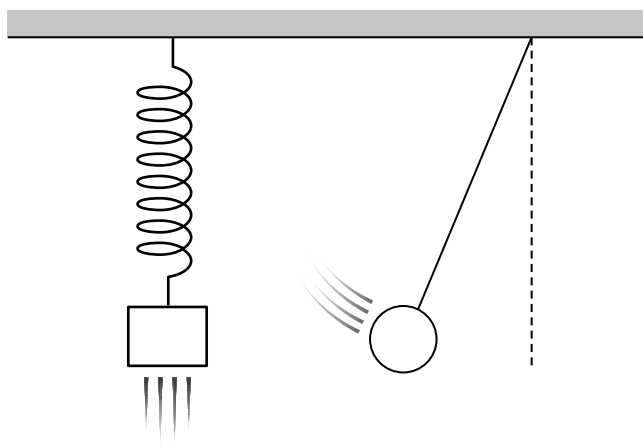
10. Block A is initially held at rest on a horizontal table and is connected to a horizontal spring that is fixed to a wall. The spring is initially at its relaxed length. Block A is connected to Block B by a string passing over a pulley, as shown. There is friction between Block A and the table. When Block A is released, Block B moves downward, and Block A moves to the right along the surface of the table until the system returns to rest. How do the mechanical energies of the two-block – spring system and the two-block – spring – Earth system change from when Block A is held at rest to when the blocks return to rest?

| Two-Block-Spring System | Two-Block-Spring-Earth System |
|-------------------------|-------------------------------|
| (A) Increases           | Decreases                     |
| (B) Decreases           | Increases                     |
| (C) Remains Constant    | Decreases                     |
| (D) Remains Constant    | Remains Constant              |

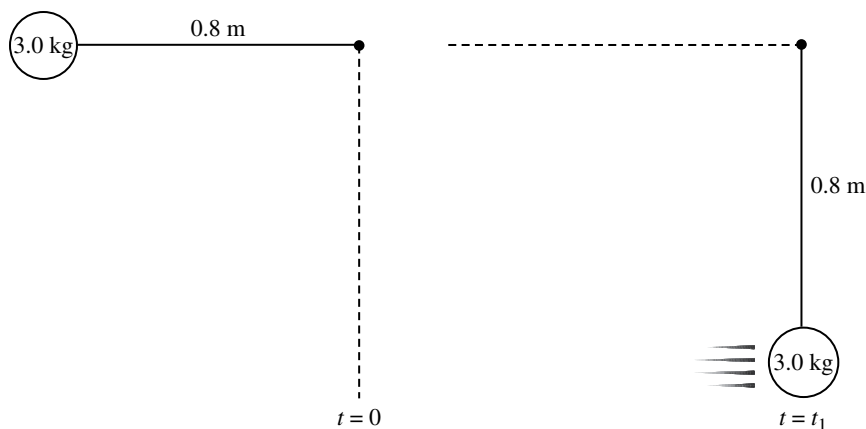
(A) A  
(B) B  
(C) C  
(D) D



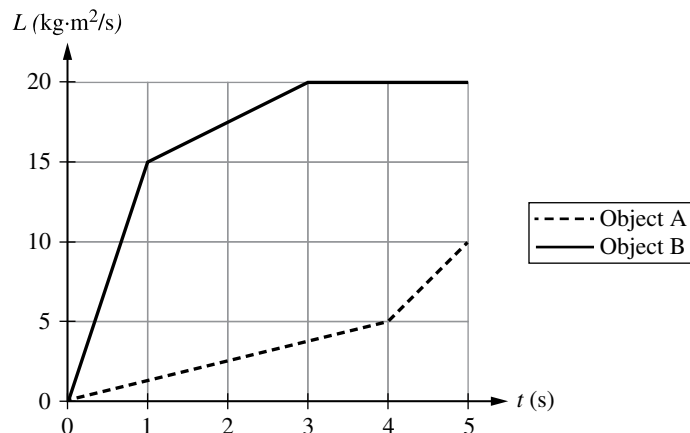
11. Block 1 slides across a horizontal surface where friction between Block 1 and the surface is negligible. Block 2 is then carefully dropped from a small height above Block 1, as shown in the figure. Block 2 sticks to Block 1 and the two-block system continues to slide horizontally together. Which of the following correctly describes the change in speed of Block 1 immediately after Block 2 sticks to Block 1?
- (A) The speed of Block 1 is less because of conservation of mechanical energy.
  - (B) The speed of Block 1 is less because of conservation of linear momentum.
  - (C) The speed of Block 1 is greater because of the conservation of mechanical energy.
  - (D) The speed of Block 1 is greater because of the conservation of linear momentum.



12. Two systems, a simple pendulum and a mass hanging on an ideal spring, both have a period of 1 s when set into small oscillatory motion on Earth. They are then set into small oscillatory motion on Planet X, which has the same diameter as Earth but twice the mass. Which of the following statements is true about the periods of the two objects on Planet X compared to their periods on Earth?
- (A) Both periods of oscillation are shorter.
  - (B) Both periods of oscillation are the same.
  - (C) The period of the mass on the spring is shorter; the period of the pendulum is the same.
  - (D) The period of the pendulum is shorter; the period of the mass on the spring is the same.

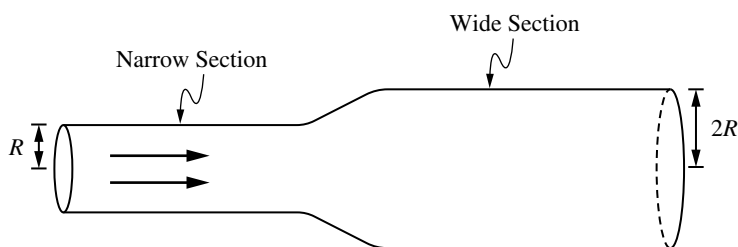


13. An object of mass  $m = 3.0$  kg is attached to one end of a string with negligible mass and length  $L = 0.80$  m. The object is released from rest at time  $t = 0$ , when the string is horizontal. At time  $t = t_1$  the object is at the location shown in the figure, where the string is vertical. Which of the following is most nearly the magnitude of the tension in the string at time  $t = t_1$ ?
- (A) 4 N  
(B) 30 N  
(C) 60 N  
(D) 90 N



14. Two disks, A and B, each experience a net external torque that varies over an interval of 5 seconds. Disk B has a rotational inertia that is twice that of Disk A. The graph shown represents the angular momentum of the two disks as functions of time between time  $t=0$  s and  $t=5$  s. The average magnitudes of the net torques exerted on disks A and B from  $t=0$  to  $t=5$  s are  $\tau_A$  and  $\tau_B$ , respectively. Which of the following expressions correctly relates the magnitudes of the average torques?

- (A)  $\tau_B = 4\tau_A$   
 (B)  $\tau_B = 2\tau_A$   
 (C)  $\tau_B = \frac{1}{2}\tau_A$   
 (D)  $\tau_B = \frac{1}{4}\tau_A$



15. An fluid flows through the two sections of cylindrical pipe shown in the figure. The narrow section of the pipe has radius  $R$  and the wide section has radius  $2R$ . What is the ratio of the fluid's speed in the wide section of pipe to its speed in the narrow section of pipe,  $\frac{v_{\text{wide}}}{v_{\text{narrow}}}$ ?

- (A)  $\frac{1}{4}$   
 (B)  $\frac{1}{2}$   
 (C) 2  
 (D) 4

## Section II: Free-Response Questions

### FREE-RESPONSE QUESTION: MATHEMATICAL ROUTINES

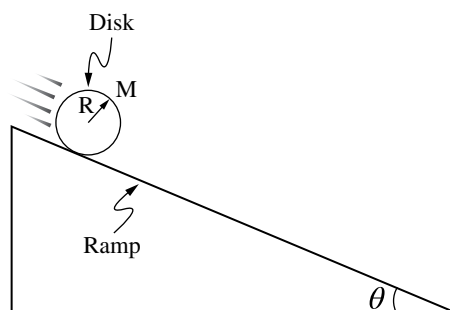


Figure 1

1. In Experiment 1, students release a disk of mass  $M$  and radius  $R$  from rest at the top of a ramp that makes an angle  $\theta$  with the horizontal. The disk rolls without slipping, as shown in Figure 1. The rotational inertia of the disk is  $I_{\text{disk}} = \frac{1}{2}MR^2$ . The frictional force exerted on the disk is  $F_f$ .
- A. i. On the diagram in Figure 2, **draw** and **label** arrows that represent the forces (not components) that are exerted on the disk as it rolls down the ramp, which is indicated by the dashed line. Each force in your diagram must be represented by a distinct arrow starting on, and pointing away from, the point at which the force is exerted on the disk.

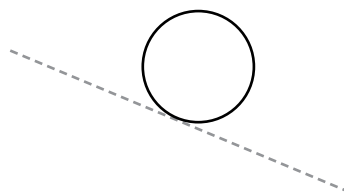


Figure 2

- ii. **Determine** an expression for the net torque exerted on the disk around the center in terms of  $\theta$ ,  $F_f$ ,  $m$ ,  $R$  and physical constants as appropriate.
- iii. **Determine** an expression for the net force exerted on the disk in terms of  $\theta$ ,  $F_f$ ,  $m$ ,  $R$  and physical constants as appropriate.
- iv. **Derive** an expression for the translational acceleration of the center of mass of the disk. Express your answer in terms of  $M$ ,  $\theta$ ,  $R$  and physical constants as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

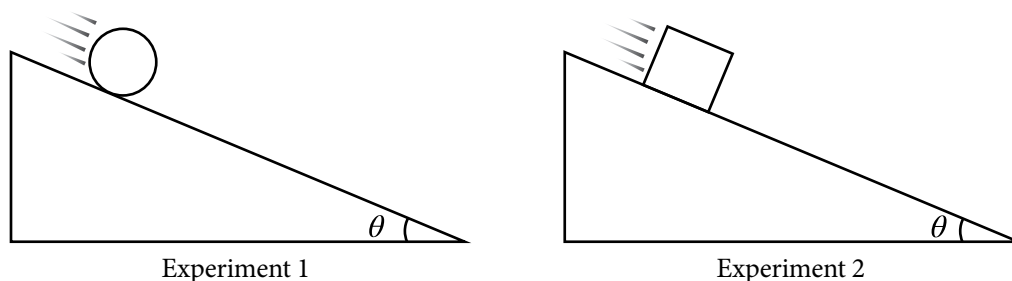


Figure 3

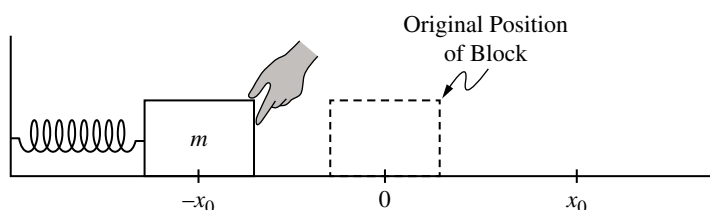
- B. In Experiment 1, the disk reaches the bottom of the ramp in time  $t_{\text{disk}}$ . In Experiment 2, the students release a block of ice from rest from the same height on a ramp inclined at the same angle  $\theta$ , as shown in Figure 3. Friction between the block and the ramp is negligible. The block reaches the bottom of the ramp in time  $t_{\text{block}}$ .

**Indicate** whether  $t_{\text{disk}}$  is greater than, less than, or equal to  $t_{\text{block}}$ ?

\_\_\_\_\_  $t_{\text{disk}} > t_{\text{block}}$       \_\_\_\_\_  $t_{\text{disk}} = t_{\text{block}}$       \_\_\_\_\_  $t_{\text{disk}} < t_{\text{block}}$

**Justify** your reasoning.

### FREE-RESPONSE QUESTION: TRANSLATION BETWEEN REPRESENTATIONS



2. A block of mass  $m$  is attached to an ideal spring, whose other end is fixed to a wall. The block is displaced a distance  $x_0$  to the left of the spring's equilibrium position, as shown in Figure 1. The block is then released from rest and oscillates with negligible friction along the horizontal surface. While the block is oscillating, it has a maximum speed  $v_{\text{max}}$ .
- A. The energy bar charts in Figure 2 represent the spring potential energy  $U_s$  of the block-spring system, and the kinetic energy  $K$  of the block, as the block passes through positions  $x = -x_0$ ,  $x = 0$  and  $x = +x_0$  while the block oscillates. The bar chart at  $x = -x_0$  is complete. **Draw** shaded rectangles to complete the energy bar charts in Figure 2 for positions  $x = 0$  and  $x = +x_0$ .
- Positive energy values are above the zero-energy line ("0"), and negative energy values are below the zero-energy line.
  - Shaded regions should start at the dashed line representing zero energy.
  - Represent any energy that is equal to zero with a distinct line on the zero-energy line.
  - The relative height of each shaded region should reflect the magnitude of the respective energy consistent with the scale shown.

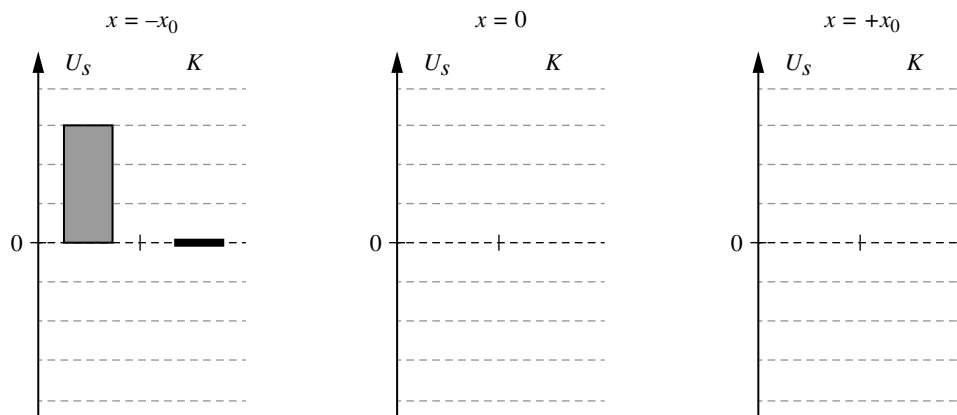


Figure 2

Figure 3 shows the position of the block as a function of time. Figure 4 shows the force exerted by the spring on the block as a function of time.

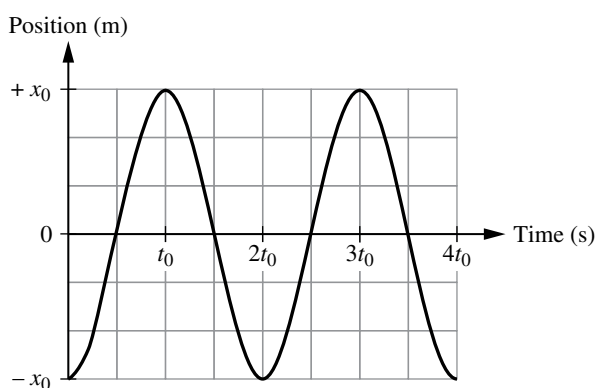


Figure 3

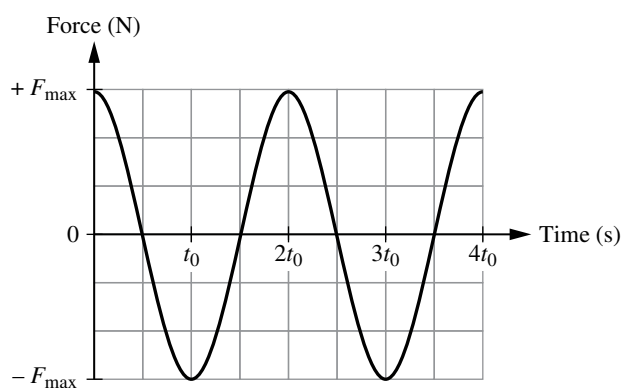


Figure 4

- B. i. Using figures 3 and 4, **determine** an expression for the spring constant of the spring.
- ii. Starting with the equation for the period of a mass on a spring, **derive** an expression for the mass of the block. Express your answer in terms of  $t_0$ ,  $x_0$ ,  $F_{\max}$ , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.
- C. On the axes provided in Figure 5 **sketch** a graph of the velocity of the block as a function of time.

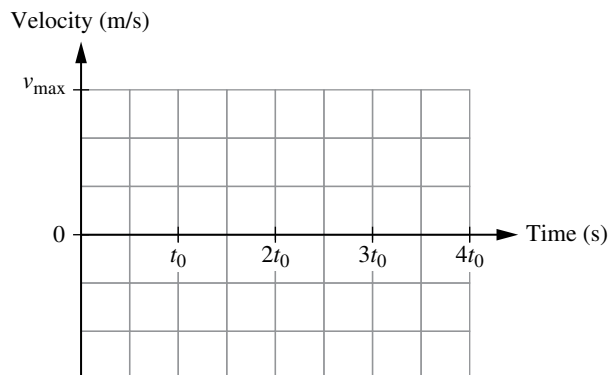


Figure 5



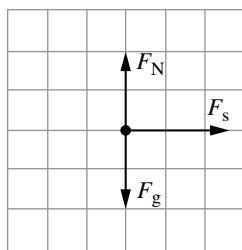


Figure 6

- D. A student sketches the free-body diagram in Figure 6, and makes the following claim:

“The free-body diagram shows the forces exerted on the block at time  $t = 1.5t_0$ ”.

**Justify** why the student’s sketch (Figure 6) and claim are or are not consistent with the graph of velocity as a function of time you sketched in part C.

### FREE-RESPONSE QUESTION: EXPERIMENTAL DESIGN AND ANALYSIS

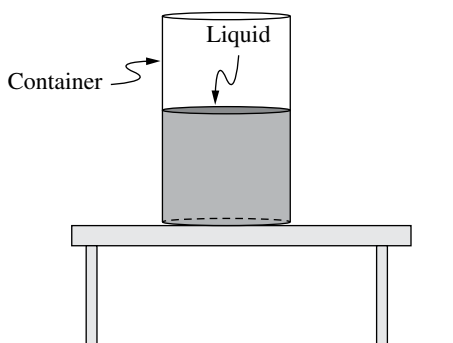


Figure 1

3. A group of students are given a cylindrical container half filled with a liquid of unknown density  $\rho$ . The students have access to an additional container with more of the same liquid, meter sticks, and a pressure sensor. They do not have access to a scale.
  - A. The students are asked to take measurements to create a graph that could be used to determine the density of the liquid. **Describe** an experimental procedure the students could use to collect the data needed to determine the density of the liquid. Include any steps necessary to reduce experimental uncertainty. If needed, you may include a simple diagram of the setup with your procedure.
  - B. **Describe** how the data collected in part A could be plotted to create a linear graph and how that graph would be analyzed to determine the density  $\rho$  of the liquid.

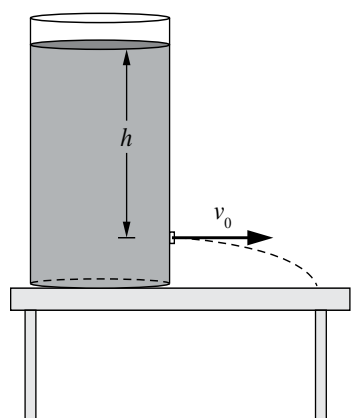


Figure 2

The students perform another experiment with another cylinder which is filled with water, as shown in Figure 2. The students make a small hole in the side of the cylinder and measure the speed  $v$  at which water exits the hole. The students plug the first hole, make another one at a different height, and repeat this procedure. Table 1 shows height  $h$  of each hole relative to the top of the water, the students made and the corresponding water speed  $v$ .

| Height $h$ (m) | Speed $v$ (m/s) |
|----------------|-----------------|
| 0.25           | 2.2             |
| 0.20           | 2.0             |
| 0.15           | 1.8             |
| 0.10           | 1.4             |
| 0.05           | 1.1             |

Table 1

- C. The students correctly determine that the relationship between  $h$  and  $v$  is given by  $v = \sqrt{2gh}$ . The students create a graph with  $v^2$  plotted on the vertical axis.
- Indicate** which measured or calculated quantity could be plotted on the horizontal axis to yield a linear graph whose slope can be used to calculate an experimental value for the acceleration due to gravity. Use the blank columns in the table to list any calculated quantities you will graph other than the data provided.  
Vertical Axis  $v^2$  Horizontal Axis \_\_\_\_\_
  - On the grid in Figure 3, **plot** the appropriate quantities to determine the acceleration due to gravity. Clearly **scale** and **label** all axes, including units, as appropriate.

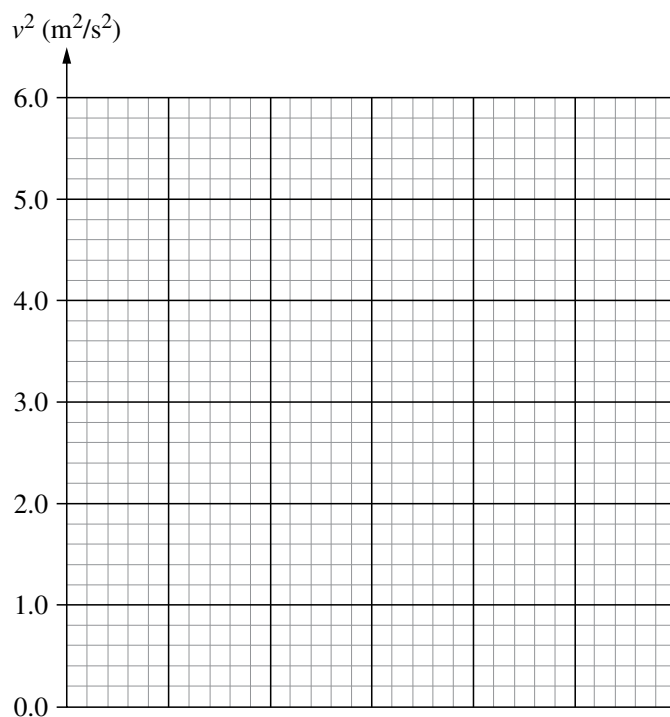


Figure 3

- iii. **Draw** a best fit line for the data graphed in part (c) ii.
- (d) **Calculate** an experimental value for the acceleration due to gravity using the best-fit line that you drew in part Ciii.

**FREE-RESPONSE QUESTION: QUALITATIVE/QUANTITATIVE TRANSLATION**



Figure 1

4. Block 1 of mass  $M_1$  slides to the right along a horizontal surface with speed  $v_0$  toward Block 2 of mass  $M_2$ , which is initially at rest, as shown in Figure 1. When Block 1 collides with Block 2, the blocks stick together. Friction between the blocks and the surface is negligible.
- A. If Block 2 is much less massive than Block 1 ( $M_2 \ll M_1$ ), **estimate** the approximate speed of the two-block system after the collision in terms of  $v_0$ . **Justify** your estimate using qualitative reasoning beyond referencing equations.
- B. Starting with conservation of momentum, **derive** an equation for  $v_f$ , the speed of the two – block system after the collision, in terms of  $M_1$ ,  $M_2$ , and  $v_0$ . Begin your derivation by writing a fundamental physics principle or an equation from the reference information.
- C. If Block 2 is much less massive than Block 1, does the equation you derived in part B agree with your qualitative reasoning from part A? **Justify** why or why not.

# Answer Key and Question Alignment to Course Framework

| Multiple Choice Question | Answer | Skill | Learning Objective | Essential Knowledge |
|--------------------------|--------|-------|--------------------|---------------------|
| 1                        | B      | 2.A   | 2.4.A              | 2.4.A.1             |
| 2                        | D      | 3.B   | 2.5.A              | 2.5.A.3             |
| 3                        | B      | 3.C   | 3.2.A              | 3.2.A.4             |
| 4                        | B      | 2.C   | 1.3.A              | 1.3.A.4             |
| 5                        | C      | 3.B   | 1.3.A              | 1.3.A.4             |
| 6                        | B      | 2.A   | 5.3.B              | 5.3.B.2             |
| 7                        | B      | 2.C   | 7.4.A              | 7.4.A.3             |
| 8                        | A      | 2.D   | 8.3.B              | 8.3.B.3             |
| 9                        | D      | 2.B   | 4.3.A              | 4.3.A.3             |
| 10                       | A      | 2.C   | 3.4.C              | 3.4.C.3             |
| 11                       | B      | 3.C   | 4.1.A              | 4.1.A.3             |
| 12                       | D      | 3.B   | 7.2.A              | 7.2.A.1             |
| 13                       | D      | 2.B   | 2.9.A              | 2.9.A.2             |
| 14                       | B      | 2.D   | 6.3.C              | 6.3.C.2             |
| 15                       | A      | 2.D   | 8.4.A              | 8.4.A.2             |

| Free-Response Question | Skill                        | Learning Objective                              |
|------------------------|------------------------------|-------------------------------------------------|
| 1                      | 1.A, 2.A, 3.B, 3.C           | 1.3.A, 2.5.A, 2.6.A, 2.7.A, 5.2.A, 5.3.B, 5.6.A |
| 2                      | 1.A, 1.C, 2.A, 2.D, 3.B, 3.C | 1.3.A, 2.8.A, 3.4.B, 7.2.A, 7.3.A, 7.4.A        |
| 3                      | 1.B, 2.B, 2.D, 3.A           | 1.5.B, 8.2.B, 8.4.B                             |
| 4                      | 2.A, 2.D, 3.B, 3.C           | 4.1.A, 4.3.A                                    |



# Scoring Guidelines

## Scoring Guidelines for Question 1: Mathematical Routines

10 points

**Learning Objectives:** 1.3.A 2.5.A 2.6.A 2.7.A 5.2.A 5.3.B 5.6.A

(a) i For a force of gravity drawn straight down from the center of the circle **1 point**

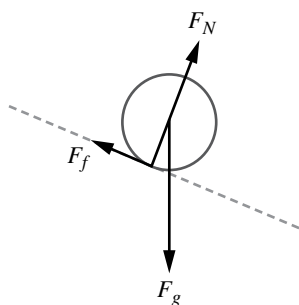
**Scoring Note:** Acceptable labels for forces include:  $F_g$ ,  $F_{\text{gravity}}$ ,  $mg$ ,  $F_{\text{earth,disk}}$ ,  $W$

$G$ , and  $g$  are not acceptable labels for the force of gravity.

For **one** of the following: **1 point**

- $F_N$  points perpendicular to the ramp starting from the point on the circle that touches the ramp.
- $F_f$  points up the ramp starting at the point on the circle that touches the ramp.

**Example Response:**



(a) ii For indicating that the torque exerted on the disk around the center of the disk is from the frictional force **1 point**

(a) iii For an expression for net force that includes a component of the gravitational force and frictional force in opposite directions.  $mg \sin \theta - F_f$  **1 point**

(a) iv For a multi-step derivation that uses Newton's second law in rotational or translational form. **1 point**

For using the relationship between the angular acceleration of the disk and the translational acceleration of the center of mass of the disk.  $\alpha = \frac{a}{R}$  **1 point**

For a correct expression for the translational acceleration of the center of mass of the disk in terms of given variables **1 point**

$$a = \frac{2}{3}g(\sin \theta)$$

**Total for part (a) 7 points**

---

**Example Response:**

$$\sum \tau = I\alpha$$

$$F_f R \sin(90^\circ) = \left(\frac{1}{2} MR^2\right) \alpha$$

$$F_f R \sin(90^\circ) = \left(\frac{1}{2} MR^2\right) \left(\frac{\alpha}{R}\right)$$

$$F_f = \frac{1}{2} Ma$$

$$\sum F = Ma$$

$$Mg \sin \theta - F_f = Ma$$

$$Mg \sin \theta - \frac{1}{2} Ma = Ma$$

$$Mg \sin \theta = \frac{3}{2} Ma$$

$$a = \frac{2}{3} g \sin \theta$$

---

**(b)** For a claim that  $t_{\text{disk}} > t_{\text{block}}$  with an attempt at a justification using forces or conservation of energy. **1 point**

For indicating that the acceleration or the net force exerted on the disk is smaller than the acceleration or the net force exerted on the block. **1 point**

**OR**

For indicating that the disk will have both rotational and translational kinetic energy at the bottom of the ramp

---

For a justification that includes that the larger acceleration for the block over the same distance leads to a smaller time for the block to reach the bottom of the incline **1 point**

**OR**

For a justification that includes that the larger translational kinetic energy for the block over the same distance leads to a higher velocity and thus a shorter time over the same distance.

---

**Example Response**

*The net force exerted on the block is larger since there is no frictional force from the surface. The net force is related to the acceleration, so since the block has a larger net force it will have a larger acceleration. Both the block and the disk start from rest and have to travel the same distance, and since the block has a larger acceleration, it will be able to travel the distance in a smaller time. Therefore, the block will reach the bottom in less time.*

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**Total for part (b) 3 points**

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**Total for question 1 10 points**

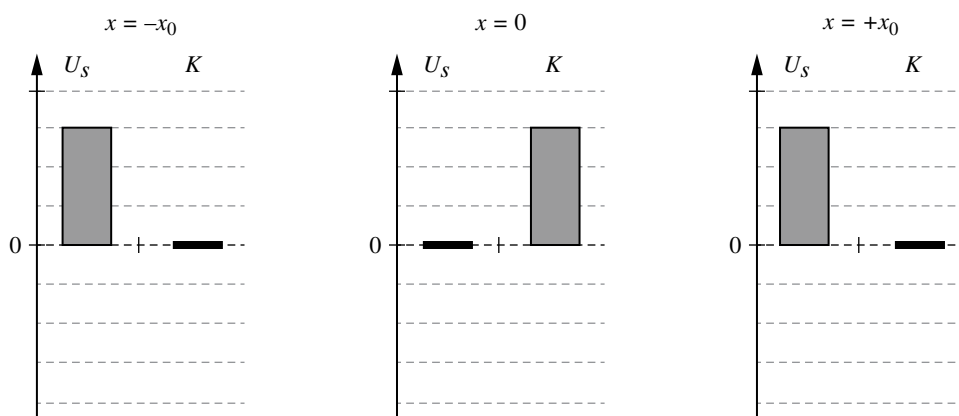
## Scoring Guidelines for Question 2: Translation Between Representations

12 points

**Learning Objectives:** 1.3.A 2.8.A 3.4.B 7.2.A 7.3.A 7.4.A

- |     |                                                                                               |         |
|-----|-----------------------------------------------------------------------------------------------|---------|
| (a) | For a single bar representing the kinetic energy of the block at $x = 0$ .                    | 1 point |
|     | For a single bar representing the potential energy of the block-spring system at $x = +x_0$ . | 1 point |
|     | For an equal total energy of the block-spring system at all three locations.                  | 1 point |

### Example Response



**Total for part (a)** 3 points

- |       |                                                                                                  |         |
|-------|--------------------------------------------------------------------------------------------------|---------|
| (b) i | For the correct spring constant in terms of $F_{\max}$ and $x_0$ .<br>$k = \frac{F_{\max}}{x_0}$ | 1 point |
|-------|--------------------------------------------------------------------------------------------------|---------|

- |        |                                                                                                                             |         |
|--------|-----------------------------------------------------------------------------------------------------------------------------|---------|
| (b) ii | For a multi-step derivation that begins with the equation for period of a mass on a spring.<br>$T = 2\pi\sqrt{\frac{m}{k}}$ | 1 point |
|--------|-----------------------------------------------------------------------------------------------------------------------------|---------|

- |                                                                                                                                                                                                                                                                           |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| For <b>ONE</b> of the following:                                                                                                                                                                                                                                          | 1 point |
| <ul style="list-style-type: none"> <li>• Substitution of the spring constant into the equation for the period of a block-spring system.</li> <li>• Substitution of <math>2t_0</math> for the period into the equation for the period of a block-spring system.</li> </ul> |         |

- |                                                                                                                                             |         |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------|
| For a correct expression of the mass of the block in terms of given variables.<br>$m = \left(\frac{t_0}{\pi}\right)^2 \frac{F_{\max}}{x_0}$ | 1 point |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------|



**Example Response:**

$$k = \frac{F_{\max}}{x_0}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$2t_0 = 2\pi\sqrt{\frac{m}{k}}$$

$$2t_0 = 2\pi\sqrt{\frac{m}{\frac{F_{\max}}{x_0}}}$$

$$t_0 = \pi\sqrt{\frac{x_0 m}{F_{\max}}}$$

$$\left(\frac{t_0}{\pi}\right)^2 = \frac{x_0 m}{F_{\max}}$$

$$m = \left(\frac{t_0}{\pi}\right)^2 \frac{F_{\max}}{x_0}$$

**Total for part (b)** **4 points**

**(c)** For a sketch that has the same period as the graph given in part (b) of  $2t_0$ . **1 point**

For a sketch that has the same magnitude above and below the zero. **1 point**

For a sketch that resembles a positive sine function. **1 point**

**Example Response**

Velocity (m/s)

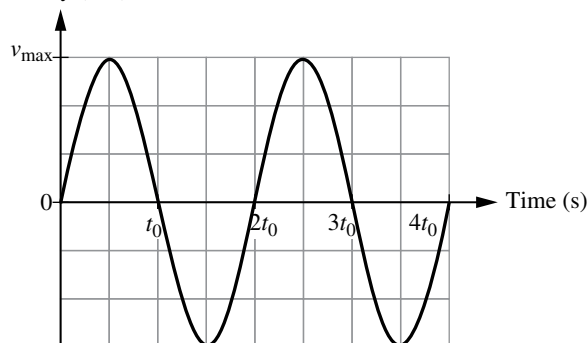


Figure 4

**Total for part (c)** **3 points**

**(d)** For a claim that the sketch and claim are or are not consistent with the graph of velocity as a function of time, consistent with part (c) with an attempt at a justification using functional dependence. **1 point**

For a justification that includes that at  $t = 1.5t_0$ , the net force should be zero with justification using a relevant feature of the graph (i.e., the slope of the graph of velocity as a function of time is zero at  $t = 1.5t_0$ ). **1 point**

---

**Example Response:**

*The claim and free-body diagram are not consistent with my graph of velocity as a function of time. At time  $t = 1.5t_0$ , my velocity graph is at a maximum, and the slope of the velocity graph at that time is zero. This means that at  $t = 1.5t_0$ , the acceleration is zero, and because acceleration is directly related to force, the force from the spring should be zero at  $t = 1.5t_0$ .*

**Scoring Note:** The two points for Part (d) can be earned independently of the rest of the points on the question, as long as the student correctly interprets the consistency between their answer for part (c) and the given sketch. For example, a student who earns zero points in part (c) can still earn full credit in (d) if they reason why their incorrect diagram is or is not consistent with the sketch.

---

|                    |          |
|--------------------|----------|
| Total for part (d) | 2 points |
|--------------------|----------|

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|                      |           |
|----------------------|-----------|
| Total for question 2 | 12 points |
|----------------------|-----------|

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## Scoring Guidelines for Question 3: Experimental Design and Analysis

10 points

Learning Objectives: **1.5.B** **8.2.B** **8.4.B**

(a) For indicating all the required quantities to determine the density of the fluid – Pressure and height of the fluid. **1 point**

For a reasonable method of reducing experimental uncertainty either by taking several pressure measurements at the same depth/height, or at least one pressure measurement at several depths/heights. **1 point**

### Example Response

*Place the pressure sensor at the bottom of the container of liquid. Measure the height of the liquid with a meterstick and record the reading on the pressure sensor. Repeat measurements to reduce experimental error. Put more liquid in the container and measure the new height of the liquid and the pressure from the sensor. Repeat for multiple heights of the liquid to reduce experimental uncertainty.*

**Total for part (a) 2 points**

(b) For describing a graph that would have a slope that is either directly or inversely proportional to the density of the fluid. Accept **ONE** of the following: **1 point**

- $P$  as a function of  $h$
- $h$  as a function of  $P$

For a correct description of how to use the slope to calculate density. Accept **ONE** of the following: **1 point**

- if Pressure as a function of height is graphed, slope is  $\rho g$
- if Height as a function of pressure is graphed, slope is  $\frac{1}{\rho g}$
- if  $\frac{P}{\rho}$  as a function of height is graphed, the slope is  $g$
- if Height as a function of  $\frac{P}{\rho}$  is graphed, the slope is  $\frac{1}{g}$
- if  $P$  as a function of  $\rho h$  is graphed, the slope is  $g$
- if  $\rho h$  as a function of  $P$  is graphed, the slope is  $\frac{1}{g}$

### Example Response:

*Graph Pressure as a function of height, the slope of the best fit line of the data will be equal to  $\rho g$ .*

**Total for part (b) 2 points**

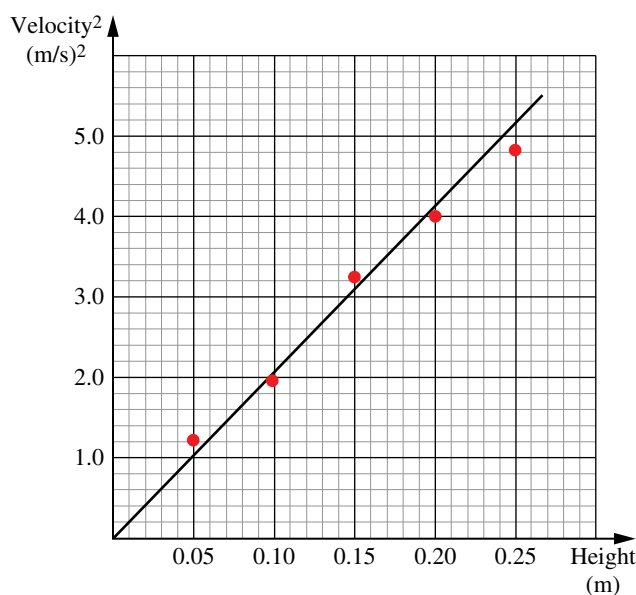
(c) i For listing appropriate quantities that could be graphed to determine the acceleration due to gravity. **1 point**

- Graphing  $v^2$  as a function of  $\Delta y$
- Graphing  $v^2$  as a function of  $2\Delta y$

**Scoring note:** Other quantities that can lead to a linear graph where the slope can be used to determine the acceleration due to gravity can earn this point.

- (c) ii For labeling axes correctly (including units) with a linear scale such that the plotted points are spread over at least half of the horizontal axis and at least half of the vertical axis. **1 point**
- Scoring Note:** This point can be earned for labels and plotted points that are consistent with the previous part.
- 
- For a graph with correctly plotted points on an appropriate scale. **1 point**
- Scoring Note:** This point can be earned for labels and plotted points that are consistent with the previous part.
- 
- (c) iii For a line that approximates the trend of the data. **1 point**
- Scoring Note:** This point can be earned for labels and plotted points that are consistent with the previous part.
- 

**Example Response:**



**Total for part (c) 4 points**

- (d) For using two points from the best-fit line to calculate the slope with appropriate units. **1 point**
- 
- For calculating the value of the acceleration due to gravity. **1 point**
- 

**Example Response:**

$$\text{slope} = \frac{1.0 - 0.0}{0.05 - 0.0} = 20 \frac{\text{m}}{\text{s}^2}$$

$$v^2 = 2gh$$

$$g = \frac{\text{slope}}{2} = \frac{20}{2} = 10 \frac{\text{m}}{\text{s}^2}$$

**Total for part (d) 2 points**

**Total for question 3 10 points**

## Scoring Guidelines for Question 4: Qualitative/Quantitative Translation

8 points

### Learning Objectives: 4.1.A 4.3.A

- (a) For a claim that the final velocity will be approximately equal to the initial velocity of Block 1. **1 point**

$$v_f \approx v_0$$

For connecting the claim to **ONE** of the following: **1 point**

- the conservation of momentum.
- Newton's third law
- the momentum of the center of mass of the system

For the justification of the evidence supporting the claim. Accept **ONE** of the following: **1 point**

- a correct statement relating either block's mass to the relative magnitude of the change in that block's momentum.
- a correct statement relating Newton's third law to a small acceleration for Block 2 or a large acceleration for Block 1
- a correct statement relating the speed of either block to the momentum of the center of mass of the system.

#### Example Response

*If the mass of Block 2 is really really small, the final momentum of Block 2 will also be really, really small. This means that the change in momentum of Block 2 is really small. The change in momentum of each block is equal and opposite, so if the change in momentum of Block 2 is small, so is the change in momentum of Block 1, so Block 1 is still traveling at approximately the same speed.*

#### OR

*Newton's third law states that the force that the two blocks exert on each other must be equal in magnitude and opposite in direction. If the mass of Block 2 is very very small, the force exerted on it by Block 1 will mean that relative to Block 1, Block 2 will have a large acceleration (changing its velocity from zero to almost  $v_0$ ), while that same force exerted on Block 1 will result in a very small acceleration – so the speed of Block 1 remains almost equal to  $v_0$ .*

#### OR

*If the mass of Block 2 is very very small, the addition of the mass of Block 2 to the mass of the system is a very small mass change. Which means that the additional momentum of adding the momentum of Block 2 moving is very very small, which means that the final speed of the system should be equal to the initial speed of the system because of the conservation of momentum.*

**Total for part (a) 3 points**

- (b) For a multistep derivation that begins with conservation of momentum **1 point**

$$p_i = p_f$$

For substituting appropriate values of mass and initial velocities **1 point**

$$M_1 v_{1,i} + M_2(0) = (M_1 + M_2) v_f$$

For a correct equation for the final velocity of the two-block system after the collision. **1 point**

$$v_f = \frac{M_1 v_0}{(M_1 + M_2)}$$

---

**Example Response:**

$$p_i = p_f$$

$$M_1 v_{1,i} + M_2 v_{2,i} = (M_1 + M_2) v_f$$

$$M_1 v_{1,i} + M_2 (0) = (M_1 + M_2) v_f$$

$$v_f = \frac{M_1 v_0}{(M_1 + M_2)}$$

---

**Total for part (b)**      **3 points**

- 
- (c) For a statement that addresses functional dependence between the mass of Block 2 and the final speed of the two-block system.      **1 point**

**Scoring Note:** It is not necessary to use functional dependence correctly to earn this point.

---

For a correct justification using functional dependence to link the claim made in part (a) to the equation derived in part (b).      **1 point**

---

**Example Response:**

*In Part (a) I claimed that if the mass of Block 2 was much less than the mass of Block 1 that the final speed of the two-block system would be approximately equal to the original speed of Block 1. In the equation I derived in part (b) since the final speed of the block is inversely related to the mass of Block 2, if the mass of Block 2 is very small, it will effectively mean that the masses in the numerator and denominator cancel leaving the final mass equal to the initial mass, which is what I claimed in part (a).*

---

**Total for part (c)**      **2 points**

---

**Total for question 4**      **8 points**